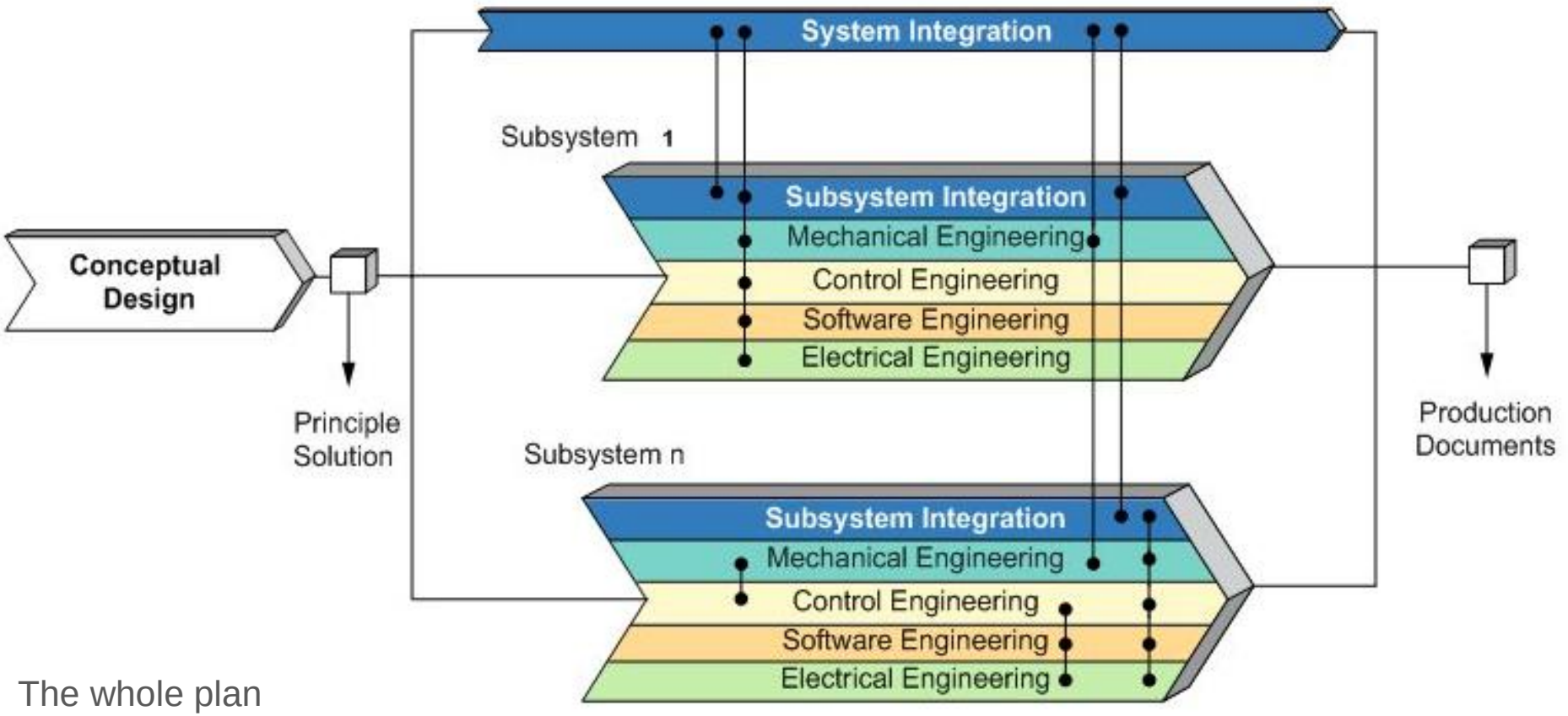
HOCHSCHULE
HAMM-LIPPSTADT

Stefan Henkler E-Mail: stefan.henkler@hshl.de

Faezeh Pasandideh E-Mail: faezeh.pasandideh@hshl.de

2 III.2 Macro Cycle

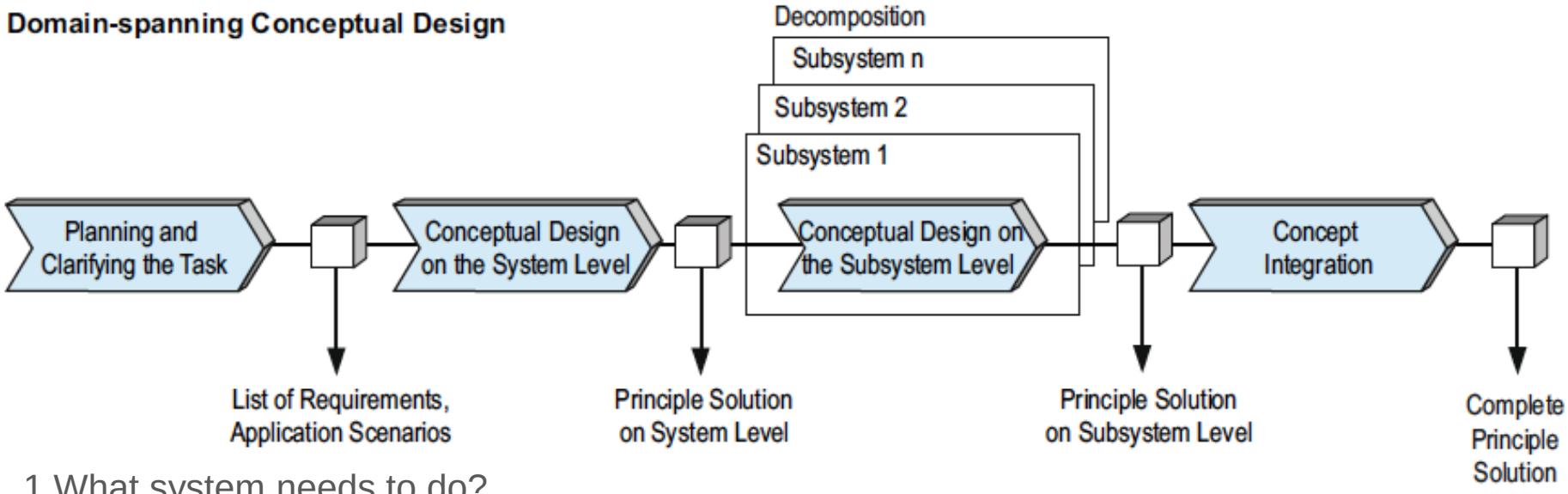


The whole plan

Big picture, a large scale lifecycle phase

3 III.3 Domain-Spanning Conceptual Design

Four main phases of the domain-spanning conceptual design



1.What system needs to do?

2. Big picture, overall system as a whole.

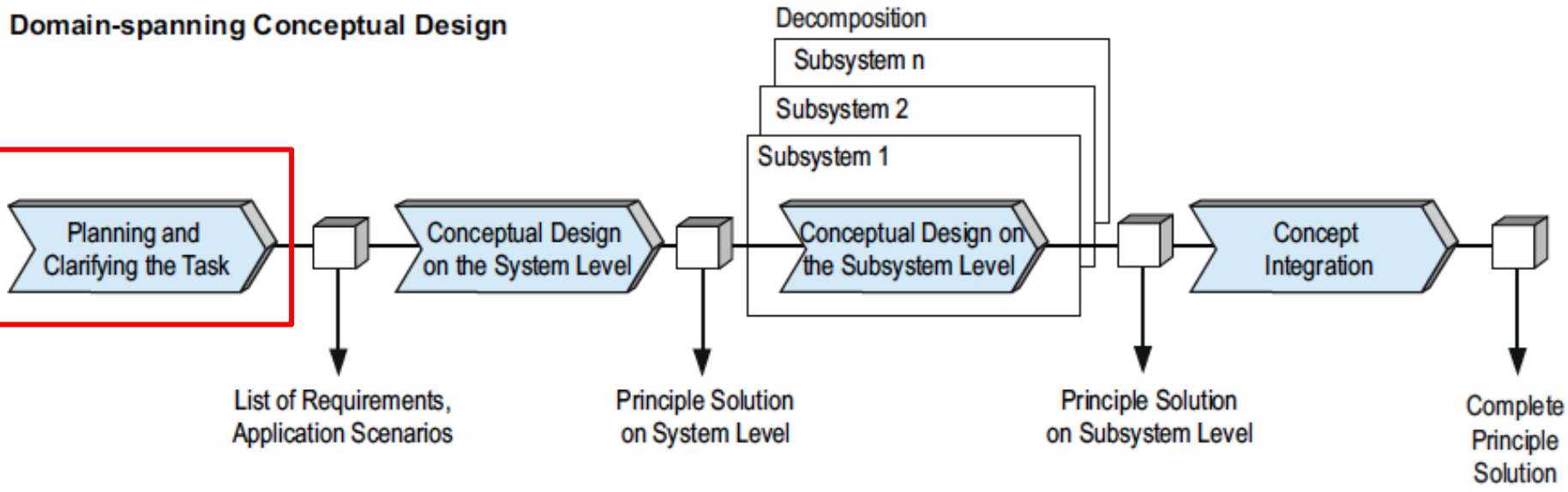
3.Decomposed to subtasks

4.Merge back into one unified solution.

How you go from an idea to a complete design concept?

4 III.3 Domain-Spanning Conceptual Design

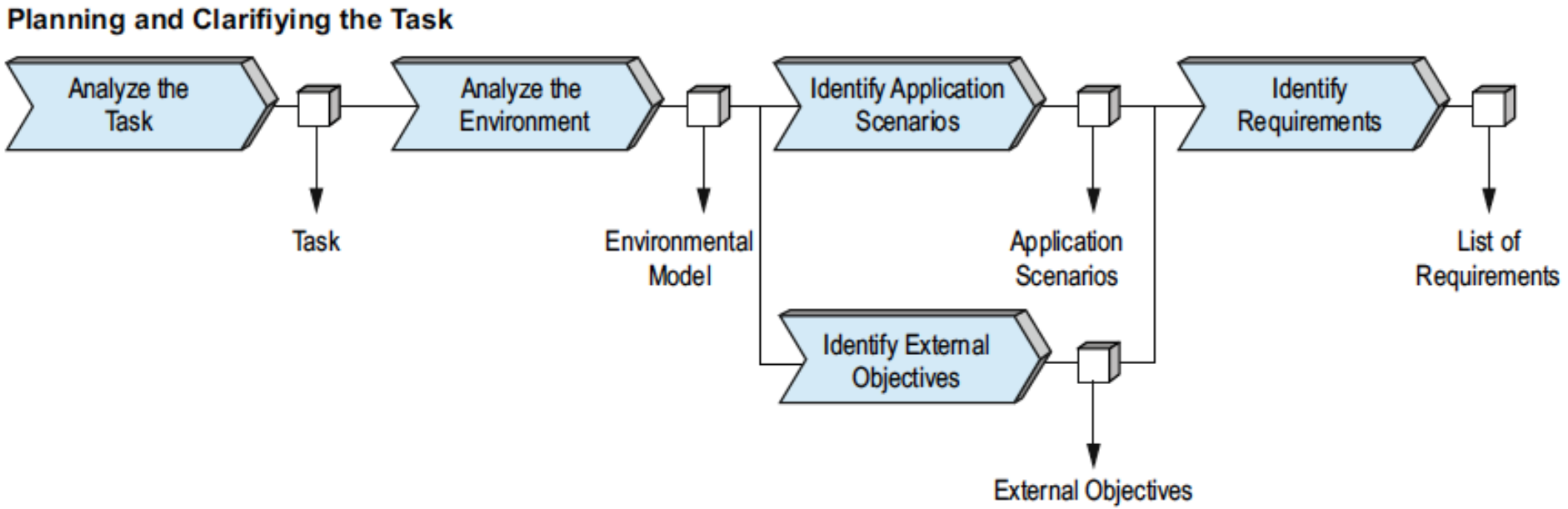
Four main phases of the domain-spanning conceptual design



1.What system needs to do?

III.3 Domain-Spanning Conceptual Design

Planning and Clarifying



1.What we're supposed to build?

2. The context system work in (physical, regulatory, organizational surroundings.

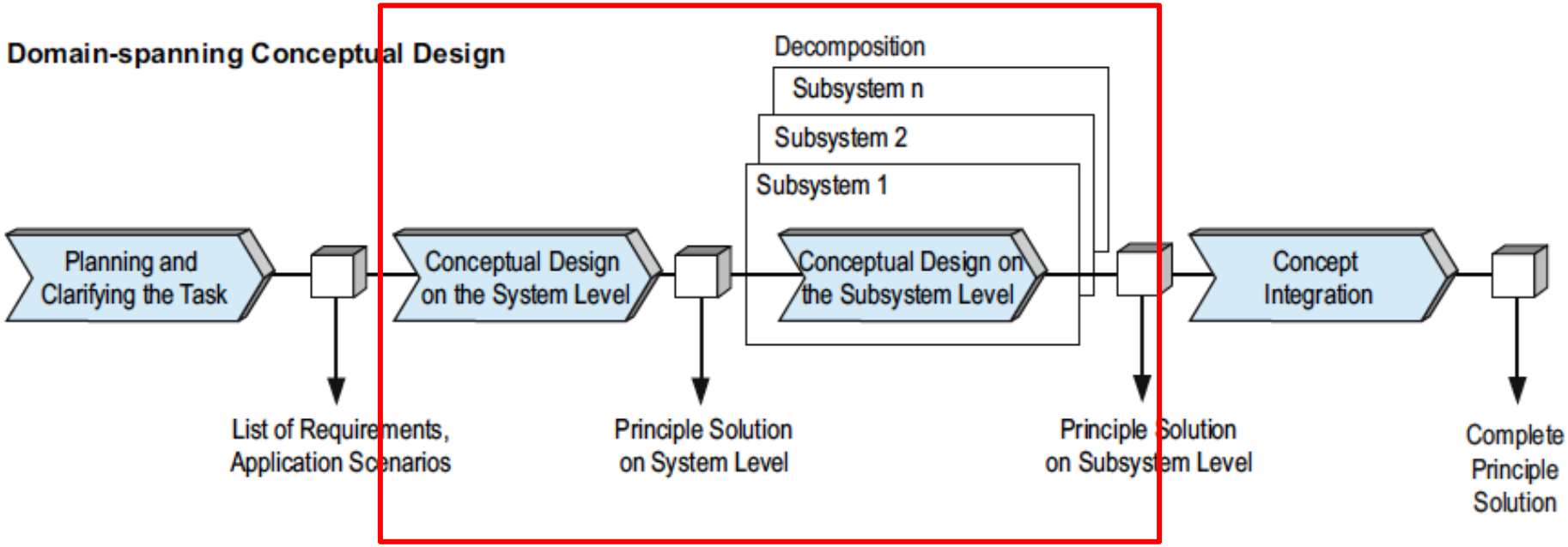
3.1. how/when system will be used

3.2. goals, constraints imposed from outside, standards, regulations, stakeholders

4. What system must do and satisfy.

III.3 Domain-Spanning Conceptual Design

Four main phases of the domain-spanning conceptual design

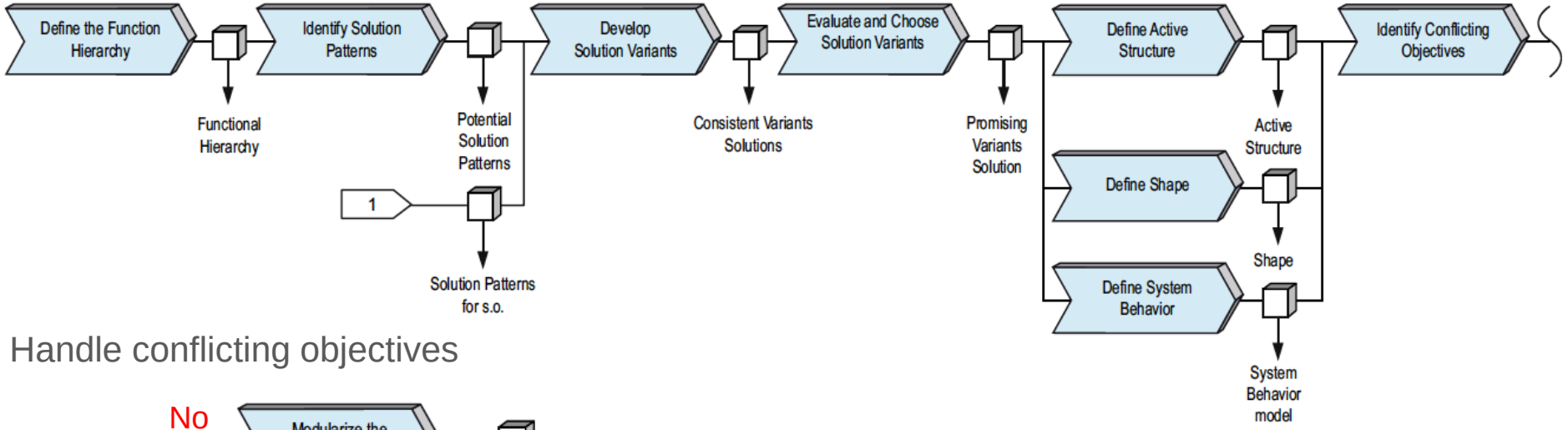


2. The context system work in (physical, regulatory, organizational surroundings.

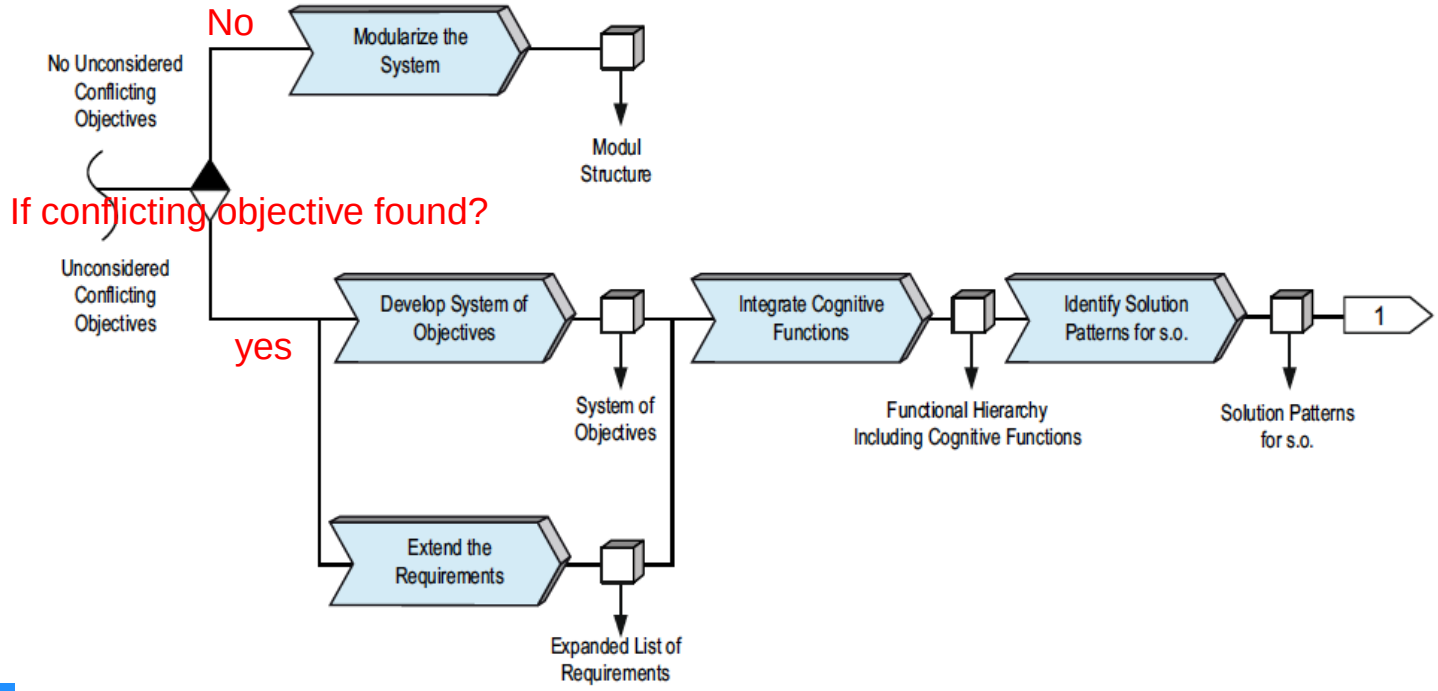
3.1. how/when system will be used

3.2. goals, constraints imposed from outside, standards, regulations, stakeholders

Conceptual Design on System Level

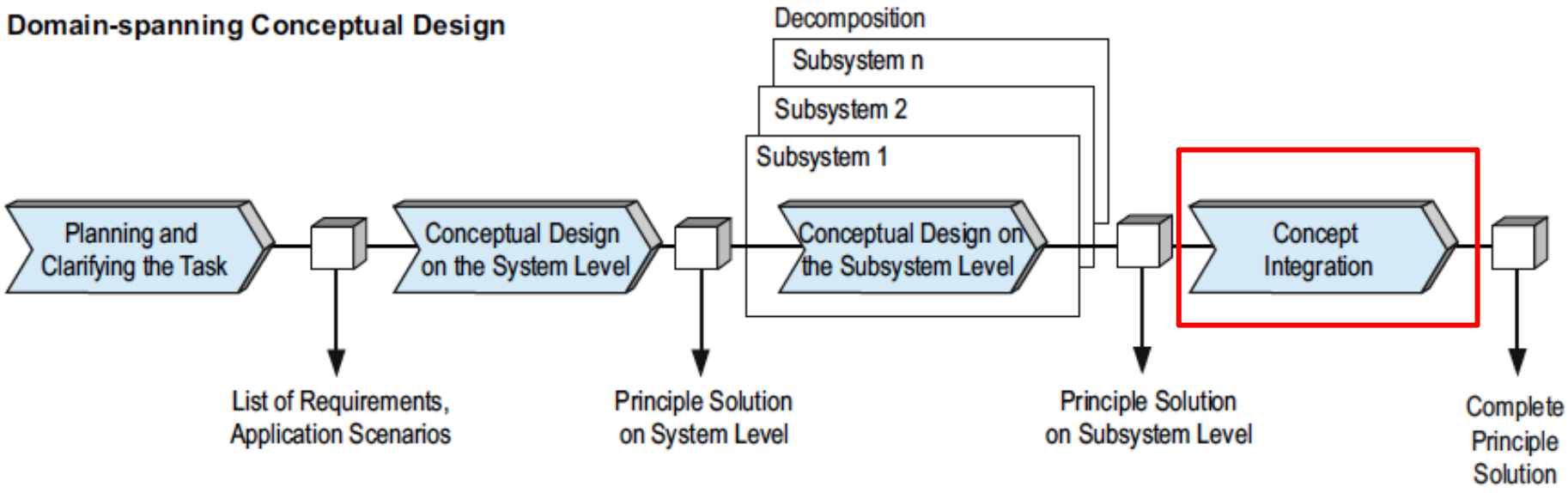


Handle conflicting objectives



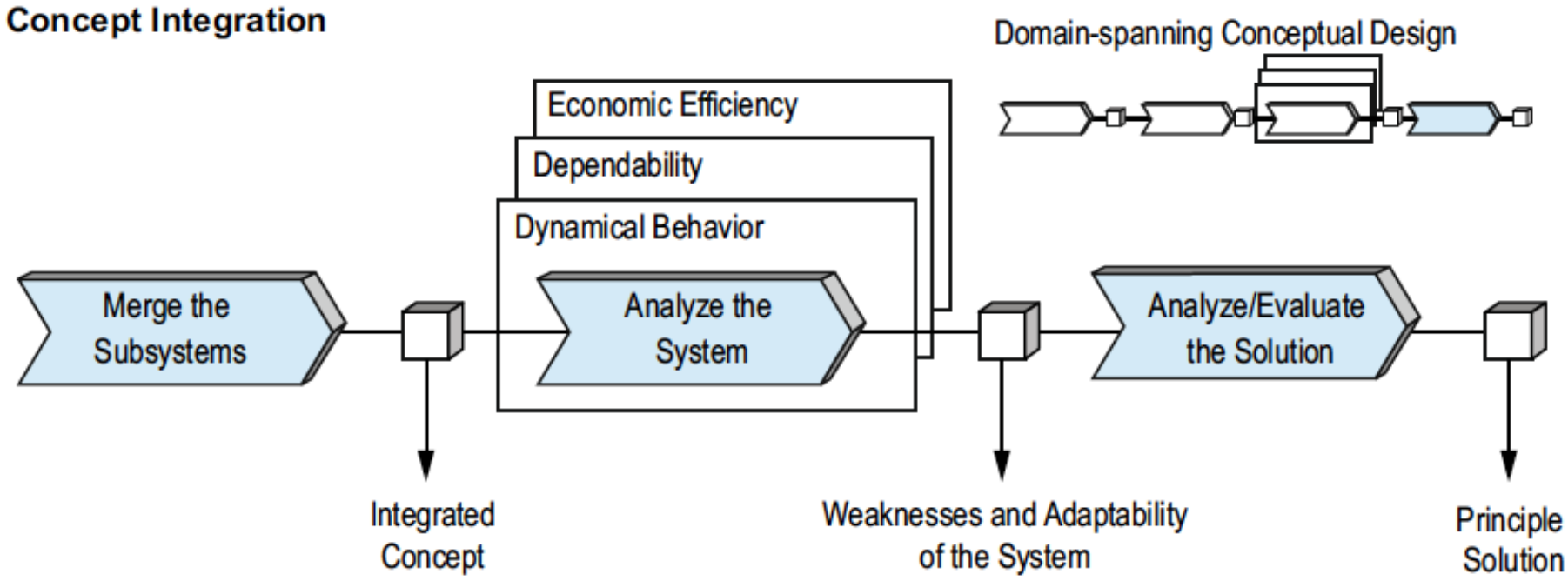
III.3 Domain-Spanning Conceptual Design

Four main phases of the domain-spanning conceptual design



4.Merge back into one unified solution.

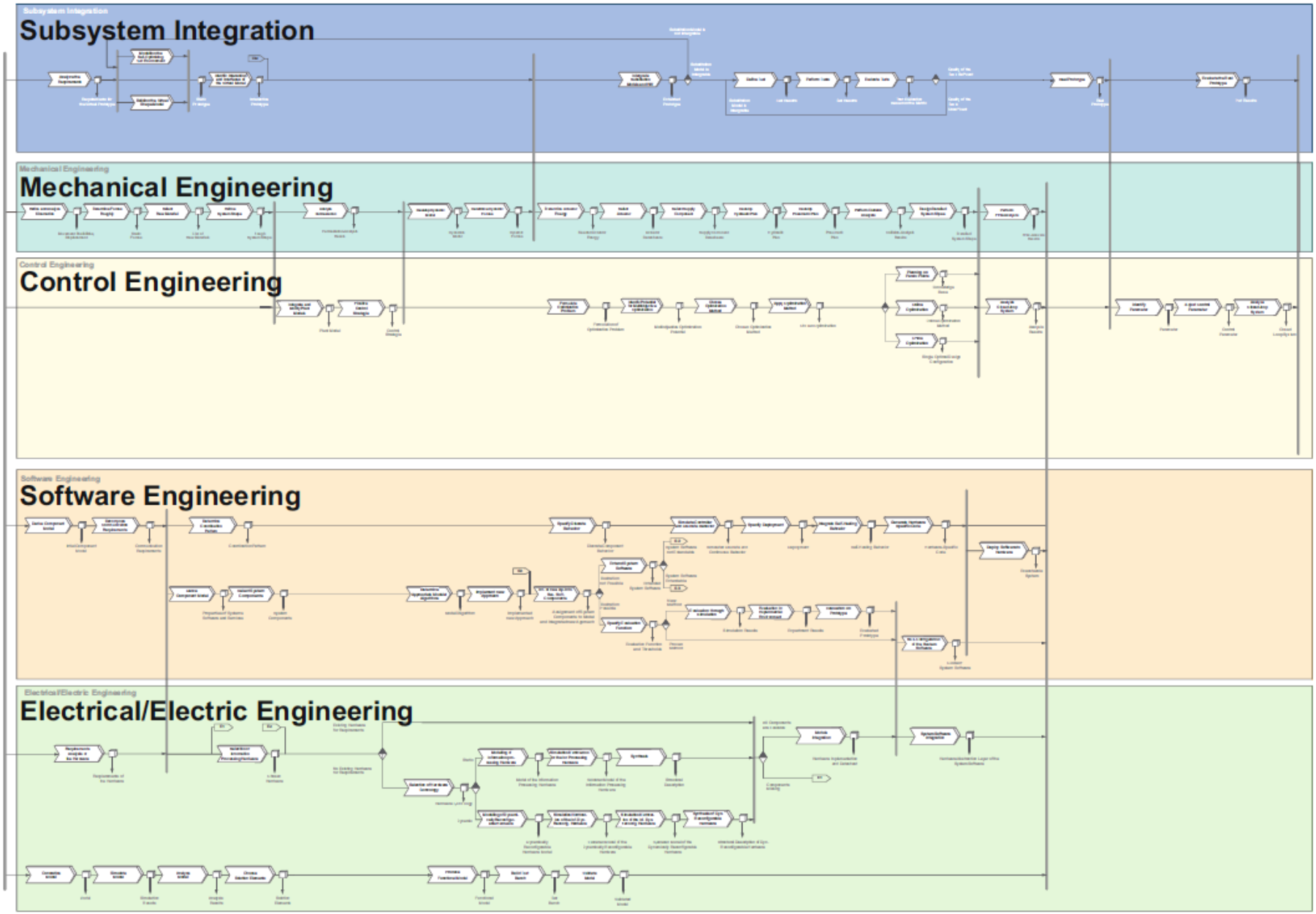
Concept Integration



1. Creates an integrated all subsystem Concept
2. Simultaneously(cost-effective?, reliable and safe?, behave correctly over time?)
3. Final output, evaluation
4. Reference: where this step sits within the overall Domain-spanning Conceptual Design flow.

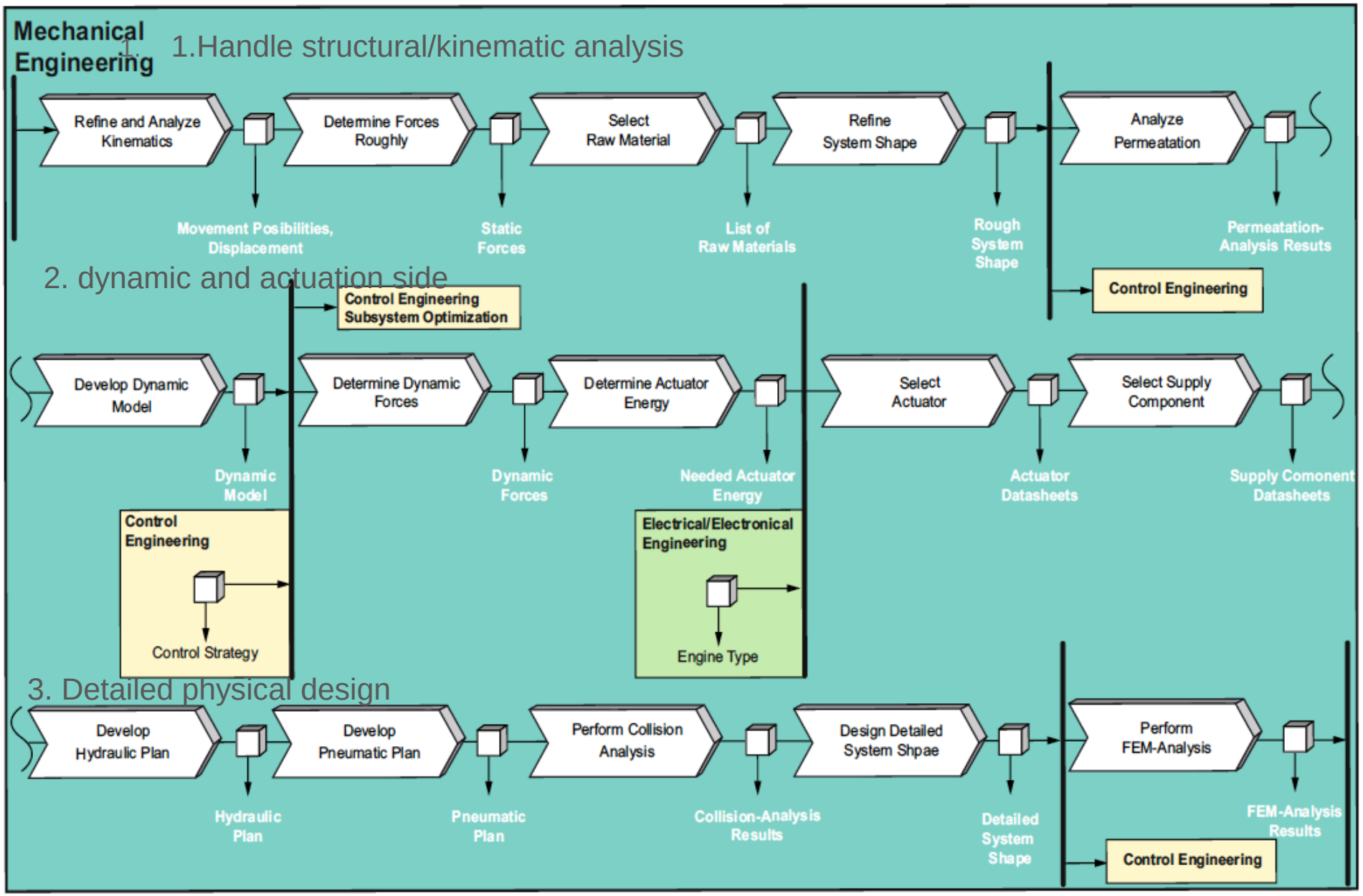
III.4 Domain-Specific Design and Development

Overview: Swimlane process diagram. Engineering disiplines, in parallel

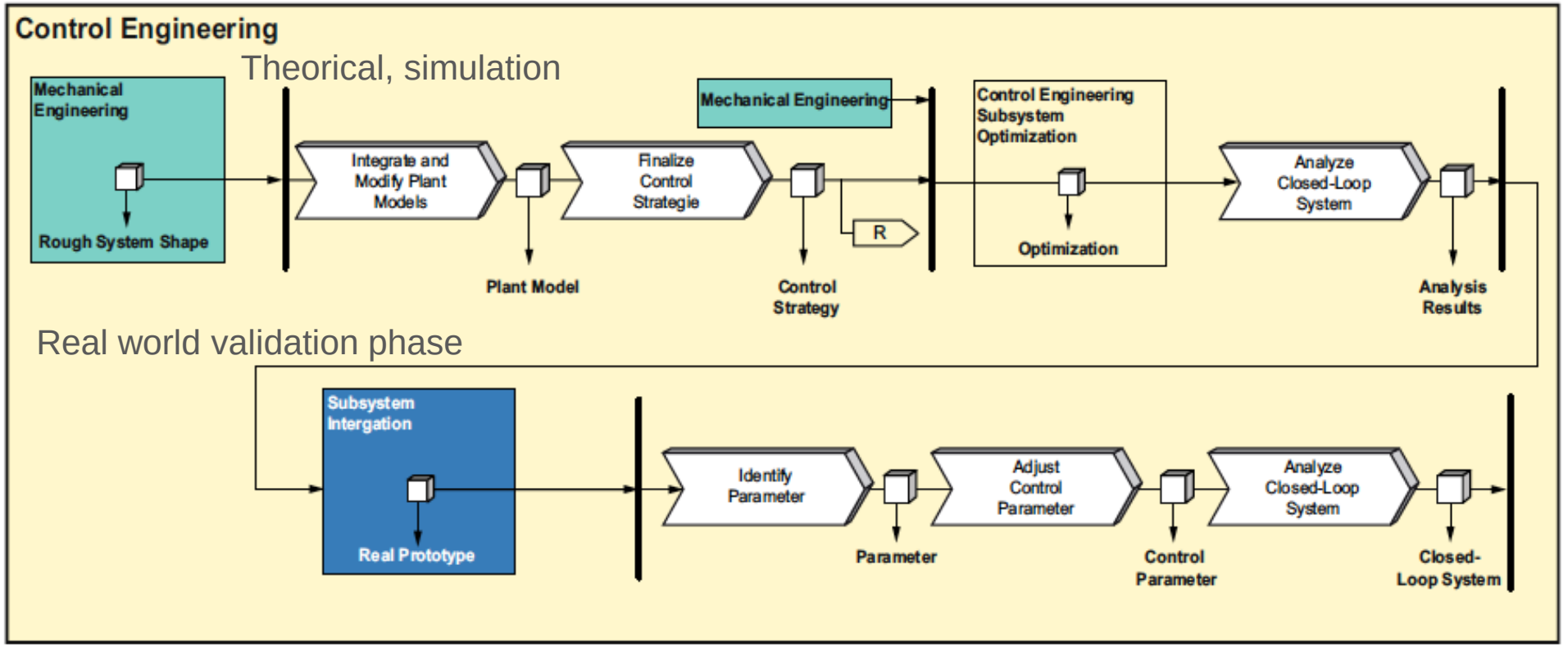


11 Domain-Specific Design and Development

Mechanical Engineering tasks broken into seq steps across multiple design layer



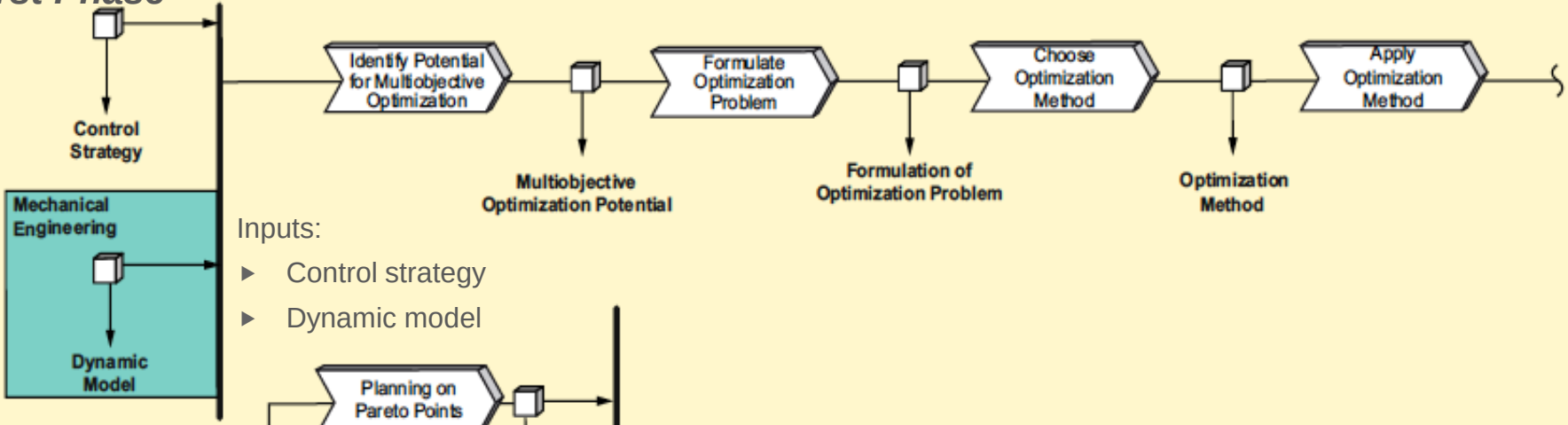
Control Engineering



Control-System Sub-System Optimization

Control Engineering Subsystem Optimization

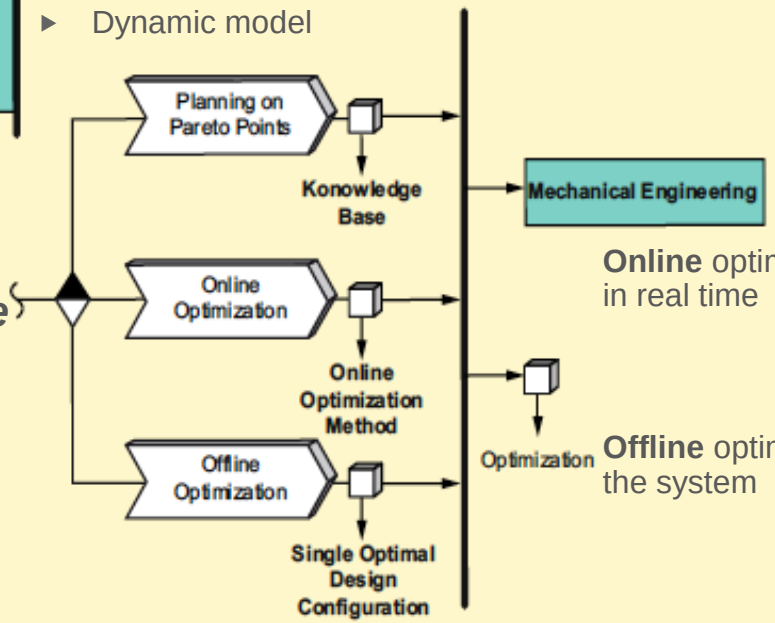
First Phase



Inputs:

- Control strategy
- Dynamic model

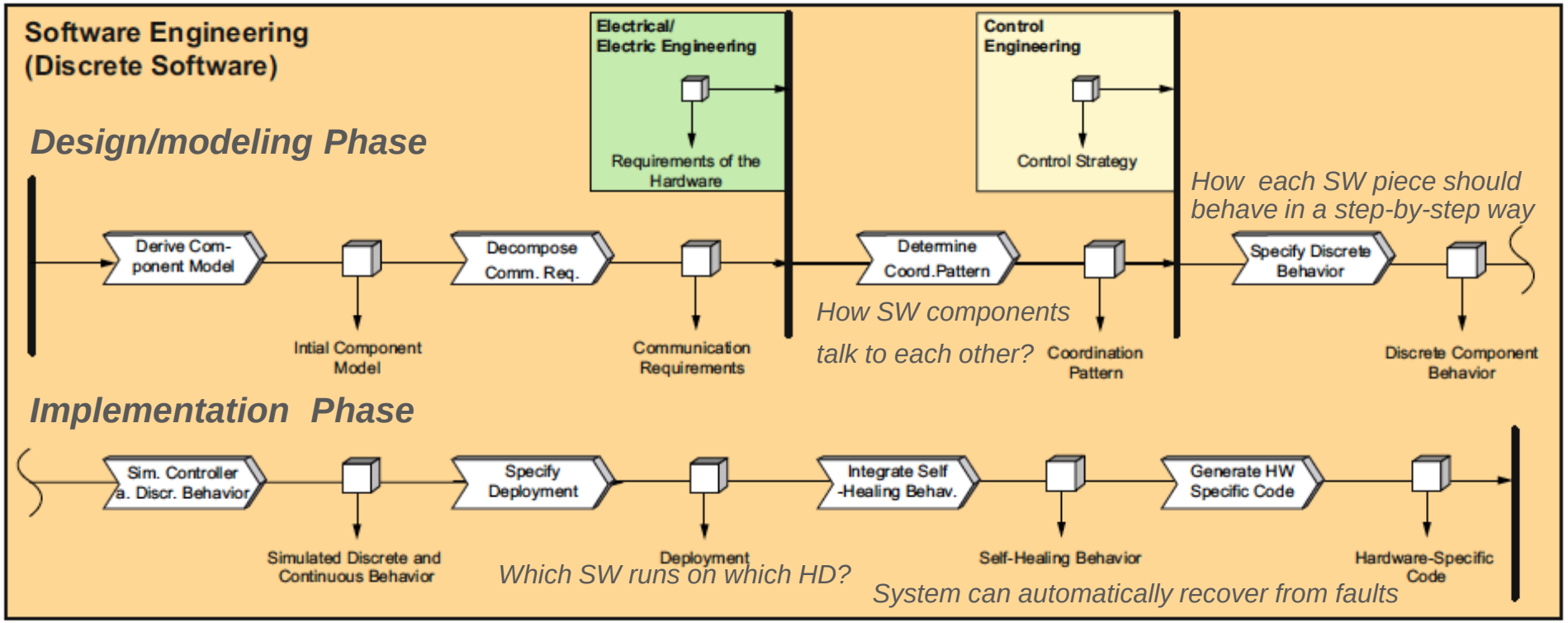
Second Phase



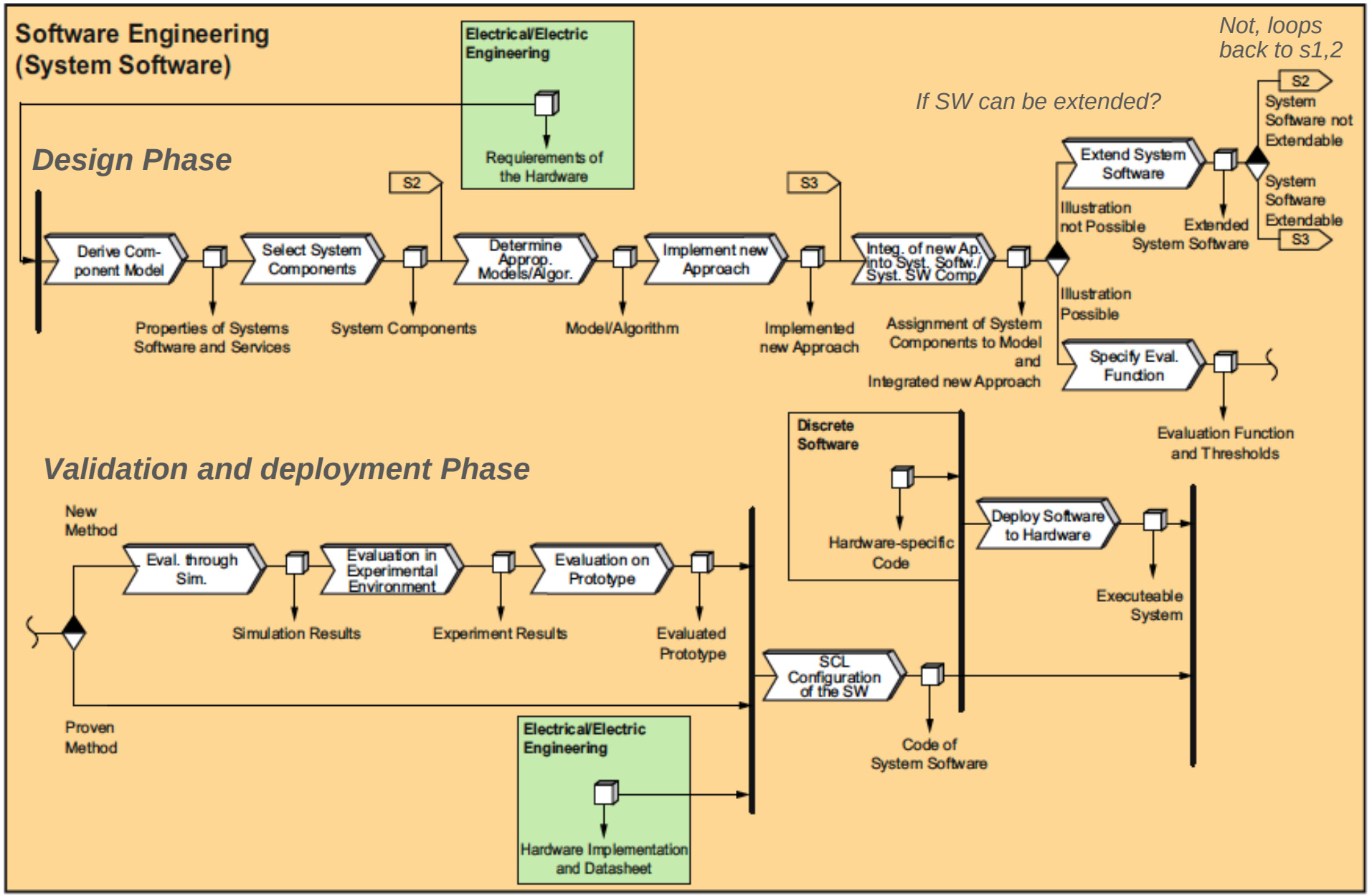
Online optimization: continuously optimizing **while** system is running in real time

Offline optimization: find one best design configuration **before** running the system

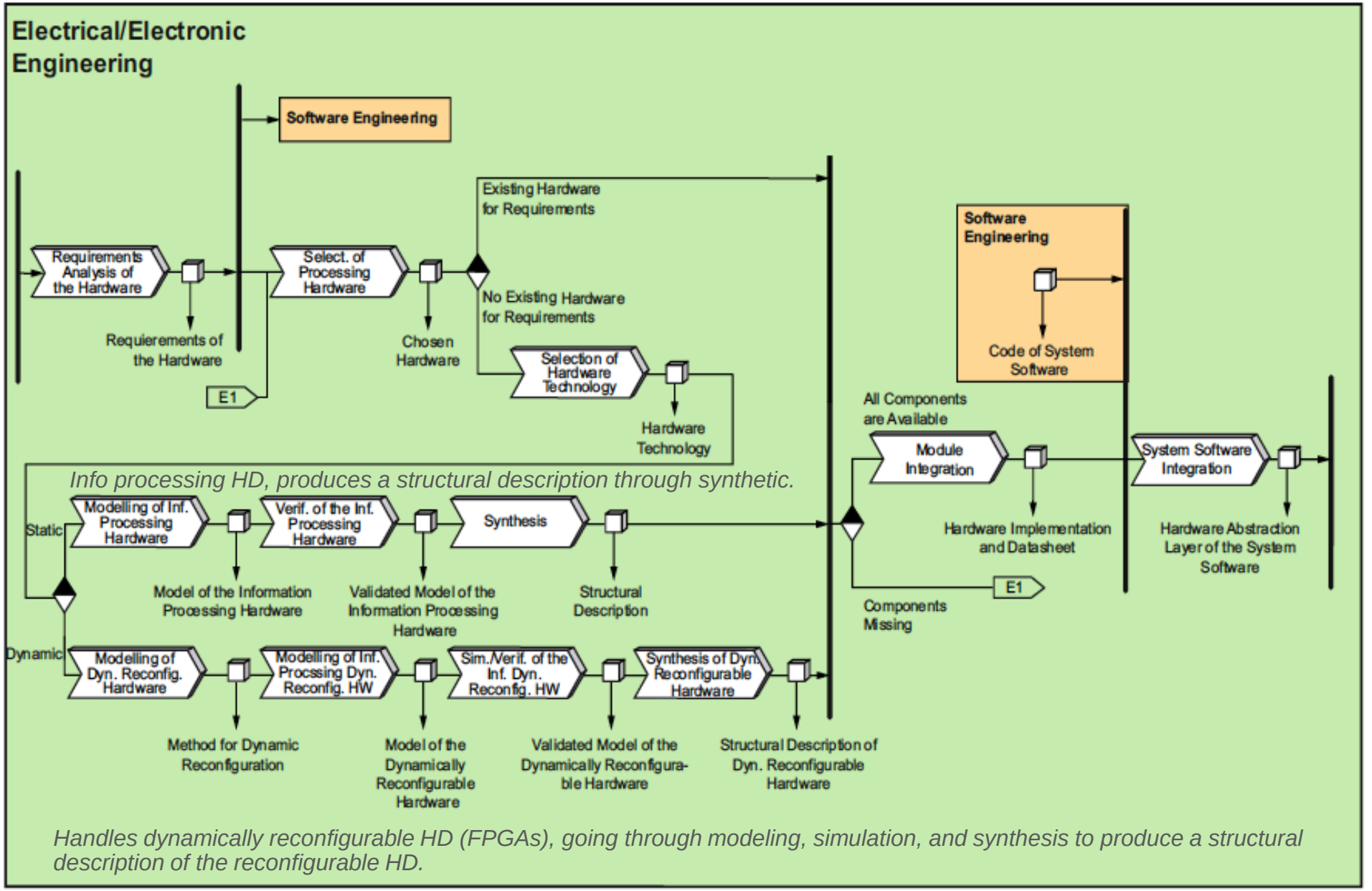
3 competing goals

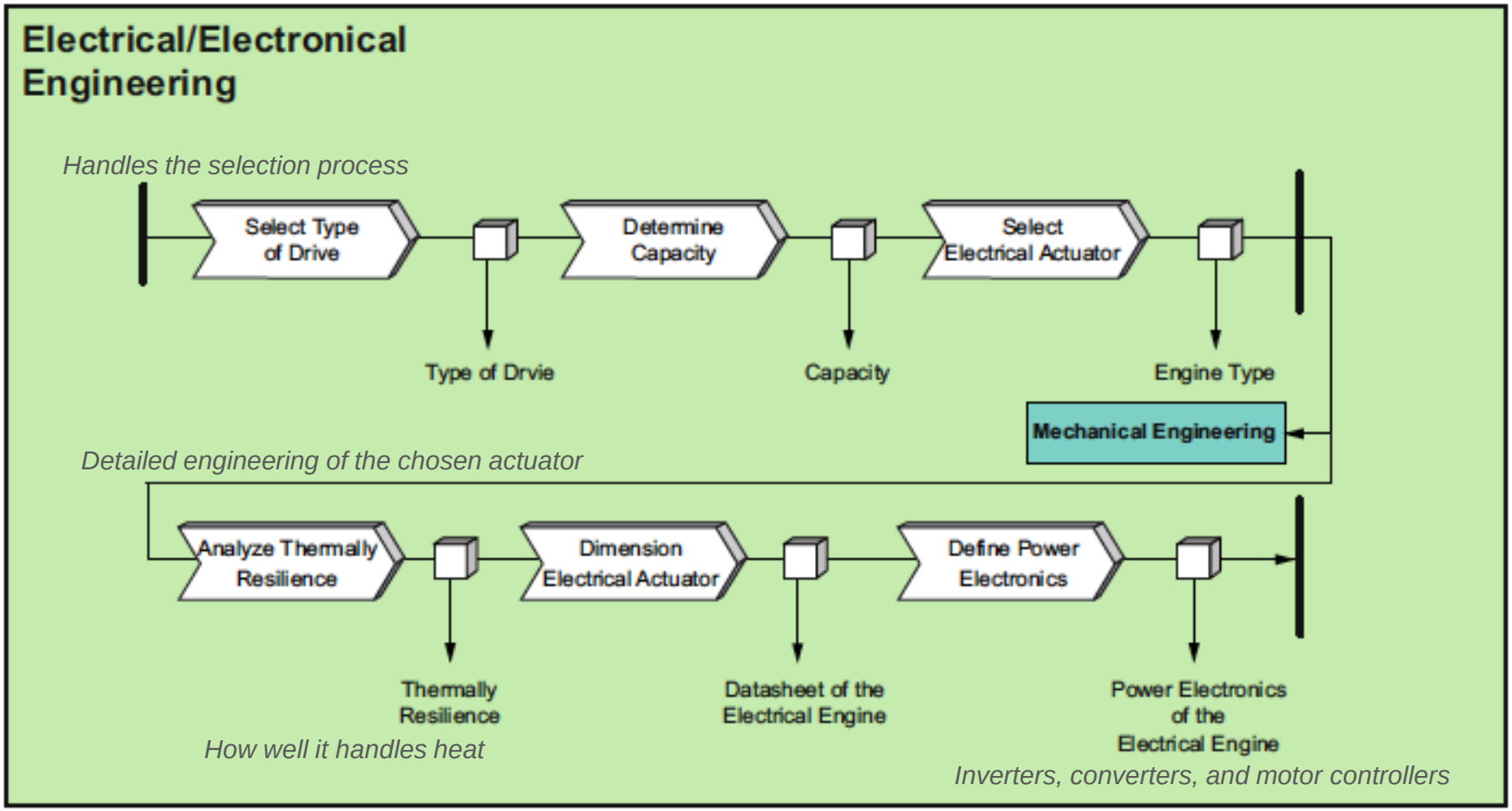


It takes system-level models and turns them into deployable, fault-tolerant embedded software code.


Software Engineering
(System Software)
Electrical/Electric Engineering
Requierements of the Hardware
Not, loops back to s1,2
If SW can be extended?
Design Phase
S2
S3
Derive Com-ponent Model
Select System Components
Determine Approp. Models/Algor.
Implement new Approach
Integ. of new Ap. into Syst. Softw./ Syst. SW Comp.
Extend System Software
S2
System Software not Extendable
System Software Extendable
S3
Illustration not Possible
Extended System Software
Illustration Possible
Properties of Systems Software and Services
System Components
Model/Algorithm
Implemented new Approach
Assignment of System Components to Model and Integrated new Approach
Specify Eval. Function
Evaluation Function and Thresholds
Validation and deployment Phase
Discrete Software
Hardware-specific Code
Deploy Software to Hardware
Executeable System
New Method
Eval. through Sim.
Evaluation in Experimental Environment
Evaluation on Prototype
Simulation Results
Experiment Results
Evaluated Prototype
SCL Configuration of the SW
Code of System Software
Proven Method
Electrical/Electric Engineering
Hardware Implementation and Datasheet

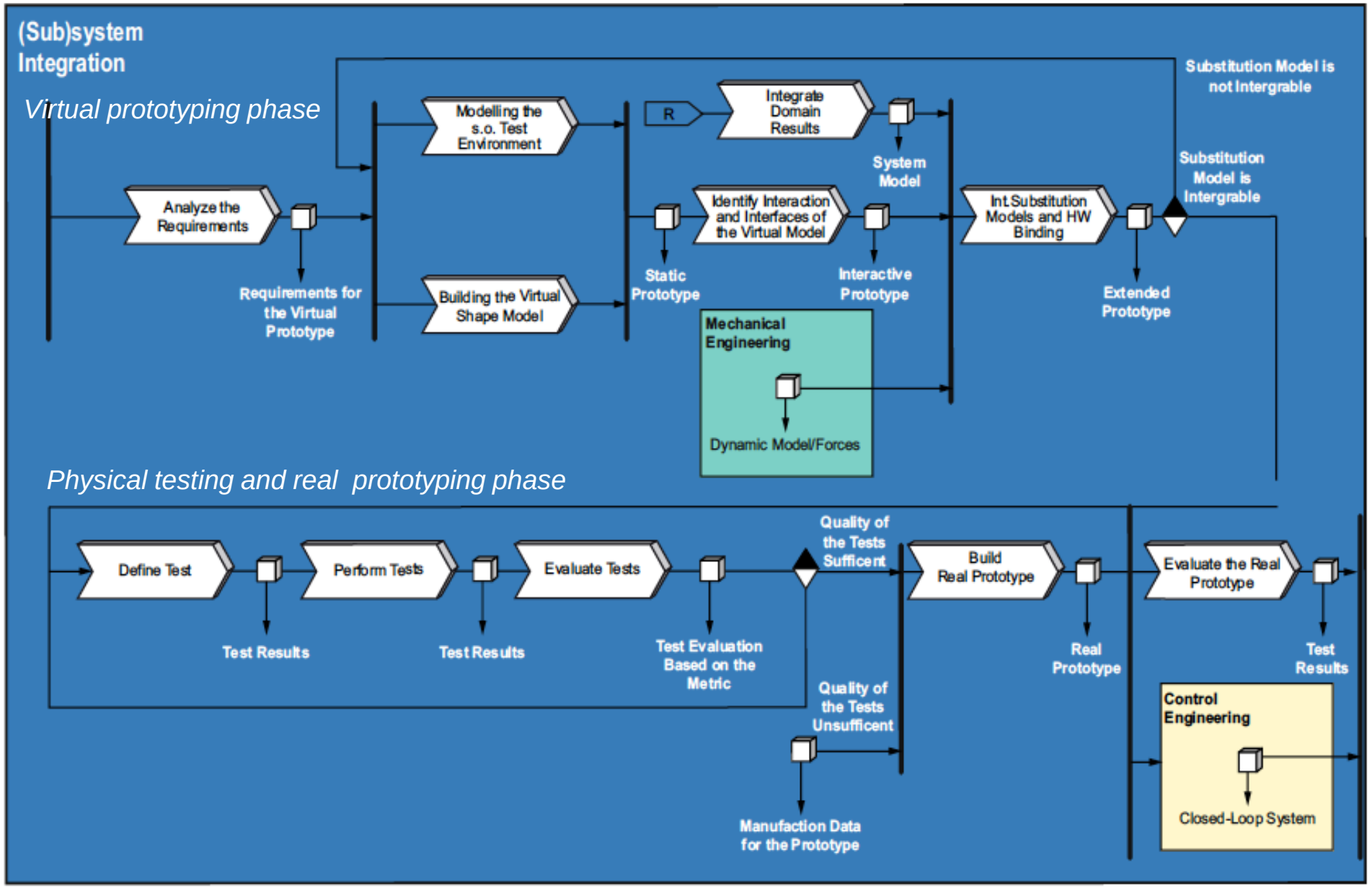
Electrical Engineering




Electrical/Electronical Engineering
Handles the selection process
Select Type of Drive
Determine Capacity
Select Electrical Actuator
Type of Drvie
Capacity
Engine Type
Mechanical Engineering
Detailed engineering of the chosen actuator
Analyze Thermally Resilience
Dimension Electrical Actuator
Define Power Electronics
Thermally Resilience
Datasheet of the Electrical Engine
Power Electronics of the Electrical Engine
How well it handles heat
Inverters, converters, and motor controllers

Domain-Specific Design and Development

Sub-System Integrateion



In the last lecture we discussed a standardized (general) systems engineering process. In this lecture we refined this process (informally) to a domain-spanning and domain-specific engineering process for the engineering of the RailCab System (as a representative for an advanced mechanical system). The explicit integration process enables the synchronization at specific domain-specific steps and it enables the engineering of advanced mechanical systems.